MILITARY SPECIFICATION SHEET

ELECTRON TUBE, TYPE 6564

This specification sheet forms a part of the latest issue of Military Specification MIL-E-1.

Description: Gas Switching. Dual T. R. Tube (Band Pass)

Application: Used with Short-Slot Hybrid Couplers

Ratings: (Absolute)	Transmitter Po kw	Open Circuit Ignitor Voltage Vdc	Altitude ft.
Minimum:	1	-700	---
Maximum:	200	---	10,000

Dimensions: See Figures 1 and 2

Pack in sealed water-vapor-proof bag. If opaque bag is used, the tube type number shall be stamped thereon.

The following tests shall be performed:

For miscellaneous requirements, see paragraph 3. 3 *in the* Inspection Instructions for Electron Tubes

Ref.	Test	Conditions	Sym	Min	Max	Unit
3.1	Qualification Approval:	Required, Note 14				
---	Marking:	Required, Note 15				
4.5	*Holding Period:*	t= 168 hours				
4.9.6	* Glass Strain:					
4.9.18.1.8	Carton Drop:	(d) Package Group 1; Carton Size K				
4.9.19.2	** Vibration:	G= 10, Note 1				
4.18.18	Voltage Standing Wave Ratio, Low Power Level:	F= 8500Mc±0.1%	δ	---	1.2	
		*F= 8800Mc±0.1%	δ	---	1.2	
		F= 9000Mc±0.1%	δ	---	1.2	
		*F= 9300Mc±0.1%	δ	---	1.2	
		F= 9600Mc±0.1%	δ	---	1.2	
		Note 2				
4.18.4.2	Duplexer Loss:	F= 8500Mc±0.1%		---	1.0	db
		*F= 8800Mc±0.1%		---	0.8	db

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Ref.	Test	Condition	Sym	Min	Max	Unit
4.18.4.2	Duplexer Loss:(Contd)					
		F = 9000Mc±0.1%		---	0.8	db
		*F - 9300Mc±0.1%		---	0.8	db
		F = 9600Mc± 0.1%		---	1.0	dn
		Ii = 100 uAdc on each ignitor, Note 3				
4.18.4.2	**Isolation	F = 8500Mc± 0.1%		15	---	db
		F - 8800Mc± 0.1%		15	---	db
		F = 9000Mc ±0.1%		15	---	db
		F= 9300Mc± 0.1%		15	---	db
		F = 9600Mc±0.1%		15	---	db
		Ii = 100 uAdc on each ignitor, Note 4				
4.18.5.1	*Ignitor Interaction:	Ii = 100 uAdc on each ignitor, Note 5		---	0.1	db
4.18.1	*Ignitor Ignition Time	Ebb = -700Vdc: R=4.0Meg: t: Note 6		---	5.0	sec
4.18.2	Ignitor Voltage Drop	Ii = 100 uAdc on each ignitor: Note 7		-200	-375	Vdc
4.18.11	Flat Leakage Power	P = 9000Mc ±0.5%Mc; pf:		---	20	mw
		pw = 40kw±10%				
		prr = 1000;				
		tpl - 1.0±0.1 us				
		tp2 = 0.5±0.05us				
		Ii = 100 uAdc on each ignitor; Note 8				

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Ref	Test	Conditions	Sym	Min	Max	Unit
4.18.10	Spike Leakage Supply	F = 9000Mc ±0.5%Mc po = 40kw ±10% prr = 1000; tpl = 1.0 ±0.1us; tp2 = 0.5 ±0.05 us; Ii = 100uAdc on each ignitor: Note 8	Ws:	---	0.1	erg
4.18.28	*Arc Loss:	F = 9000Mc ±0.5%Mc; po = 40 kw ±10% prr = 1000; tp = 1.0us ±.15 us; Note 9		---	0.8	db
4.18.17.1	Temperature Cycle:		cycles:	1	---	
4.18.17.2	Temperature Cycle:		cycles:	10	---	
4.18.15.1	*Recovery Time(1):	F = 9000Mc ±0.5%Mc; po = 200kw±10%; tp = 1.0us ±.05us; prr = 1000 Ii = 100uAdc on each ignitor; Note 10	t:	--	7.0	us

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Ref.	Test	Conditions	Sym	Min	Max	Unit
4.18.19	*Recovery Time(2):	F = 9000Mc ±0.5%Mc; po = 65kw ±10%; tp = 1.0us ±.05us; prr = 1000; Ii = 100uAdc on each ignitor; Note 10	t:	---	3.0	us
4.18.19	**High Level VSWR:	F = 9000Mc ±0.1%Mc; po = 40kw ±10%; tp = 1.0us ±.05 us; prr = 1000; Ii = 100uAdc on each ignitor: Note 11	6:	---	1.2	
4.11	Life Test	F = 9000Mc ±5%Mc; Ii = 100uAdc on each ignitor; po = 200kw±10% min.; tp = 1.0us ±.05 us; prr = 1000 Note 12; Group D	t:	500	---	hrs
4.11.4	Life Test End Point:	Recovery Time (1):	t:	---	10	us
		Recovery Time (2);	5:	---	5	us
		Flat Leakage Power;	pf:	---	20	mw
		Spike Leakage Energy;	Ws	---	.15	erg
		Duplexer Loss		---	1.0	db
		F = 9000Mc ±0.1%Mc;				

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Note 1: There shall be no evidence of shorts of the ignitor electrodes as indicated by a short indicator during vibration.

Note 2: The low level VSWR shall be measured as shown in Test Circuit A of Fib. 3. The short-slot hybrid couplers used shall have a VSWR of 1.10 maximum over the specified frequency band. Each hybrid shall split the power evenly to within C.25 db, and shall have a minimum isolation of 25 db.

Note 3: The duplexer loss shall be measured as shown in Test Circuit B of Fig. 3. The duplexer loss is defined as 10 log$\frac{\text{Power in Antenna.}}{\text{Power in Receiver Arm}}$ The adjustable short on the transmitter arm shall be adjusted for minimum power in the receiver arm of the duplexer at each frequency.

Note 4: Isolation shall be measured as shown in Test Circuit C of Fig. 4. The db isolation is defined as 10 log$\frac{\text{Power in Receiver Arm}}{\text{Power in Transmitter Arm}}$

Note 5: Ignitor interaction is measured with the specified ignitor current adjusted on both ignitor electrodes.

Note 6: Ignitor ignition time shall be measured separately for each ignitor electrode.

Note 7: Ignitor voltage drop shall be measured separately for each ignitor electrode.

Note 8: As measured in Test Circuit D of Fig. 4.

Note 9: As measured in Test Circuit E of Fig. 5.

Note 10: The measurements shall be made as shown in Test Circuit F of Fig. 5 after at least 30 minutes of operation at the condition specified.

Note 11: The high power level VSWR shall be measured as shown in Test Circuit G of Fig. 6.

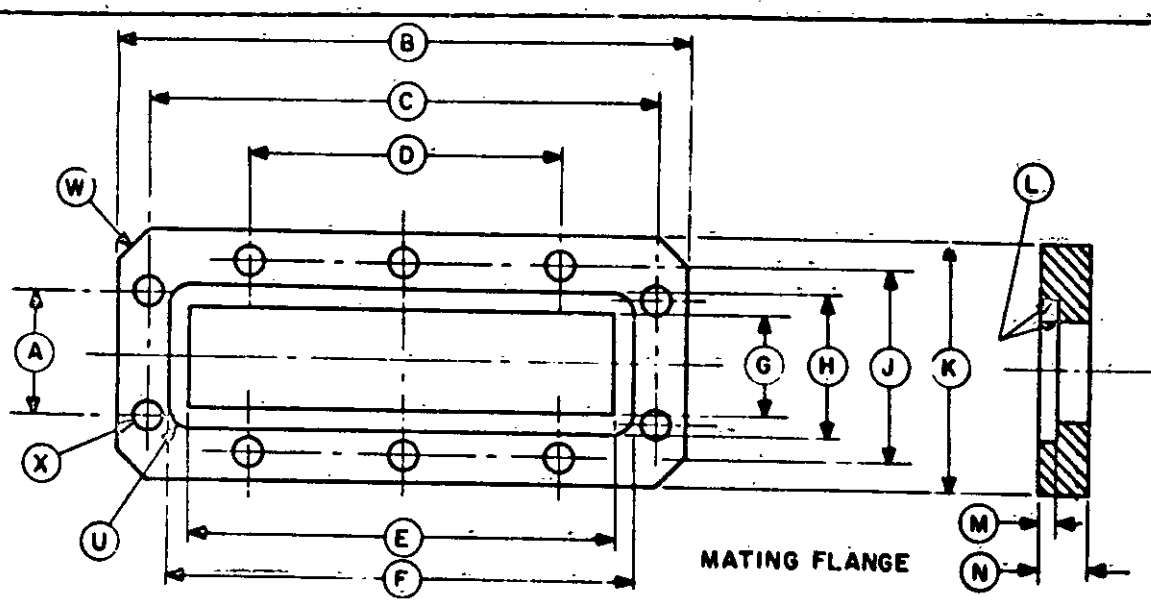
Note 12: As operated in Test Circuit H of Fig. 6.

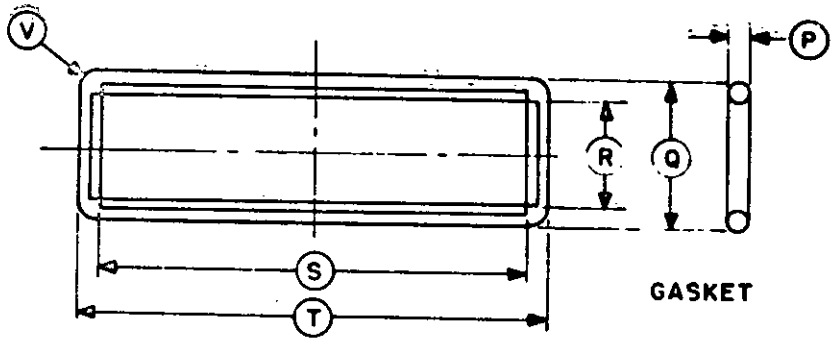
Note 13: All measurements that include short-slot hybrids shall be made with gaskets bolted between input and output flanges of tube and mating flanges.

Note 14: Application for Qualification Approval should be made in accordance with Inspection Instructions for Electron Tubes, with the exception that the request for Qualification Approval shall be addressed to Components and Materials Branch, Standardization Division, Signal Equipment Support Agency, Fort Monmouth, New Jersey.

Note 15: Electron Tube 6564, procured directly or indirectly on Army contracts, shall have "U. S. Army" marked preceding, or above, the type designation, as specified in the bid request or contract. The letters "JAN" or an abbreviation thereof shall not be used.

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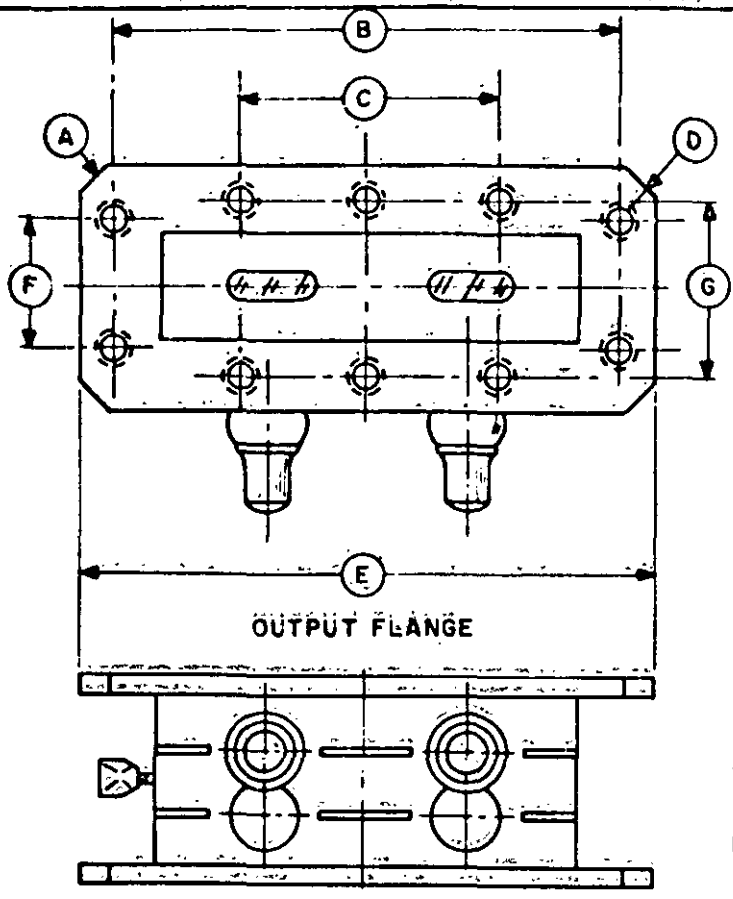




REF.	DIMENSIONS	REF.	DIMENSIONS
A	.722 ± .004	M	.069 ± .002
B	3.210	N	.250
C	2.092 ± .008	P	.110 TYP
D	1.448 ± .008	Q	.931 MAX
E	2.440 ± .002/.001	R	.630 MIN
F	2.684 ± .004	S	2.444 MIN
G	.628 ± .002/.001	T	2.647 MAX
H	.872 ± .004	U	.046 ± .002 R
J	1.060 ± .004	V	.046 ± .002 R
K	1.380	W	.120 X 45° CHAMFER OR 1-1/8 R
L	.005 R MAX	X	.171(.173) DR. (10) HOLES

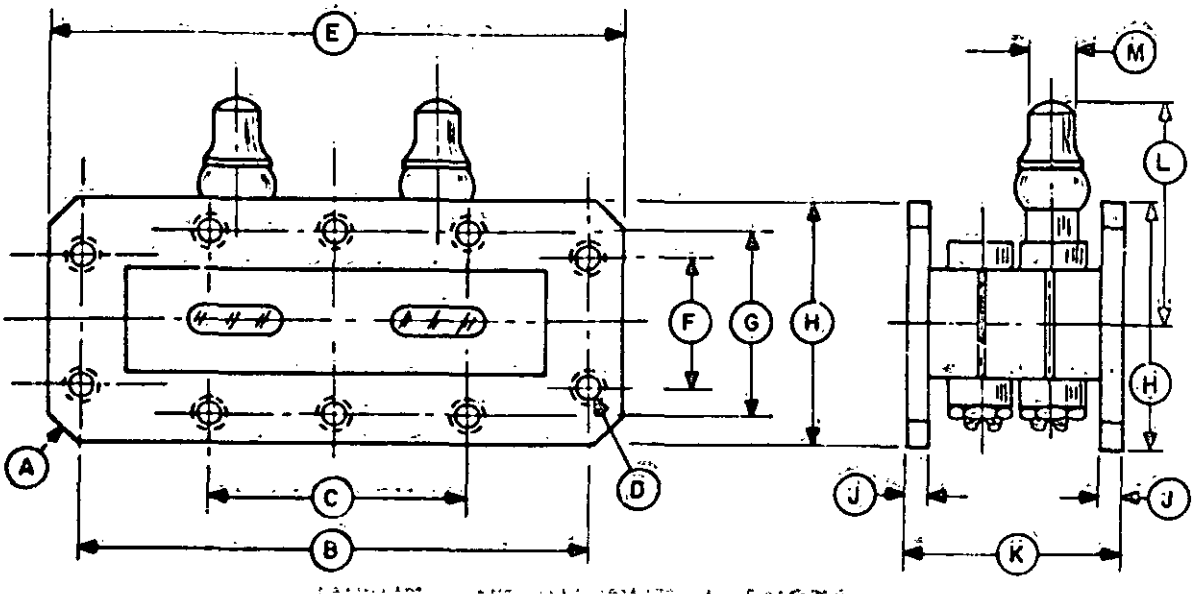
MATING FLANGE AND GASKET
FIG. 1

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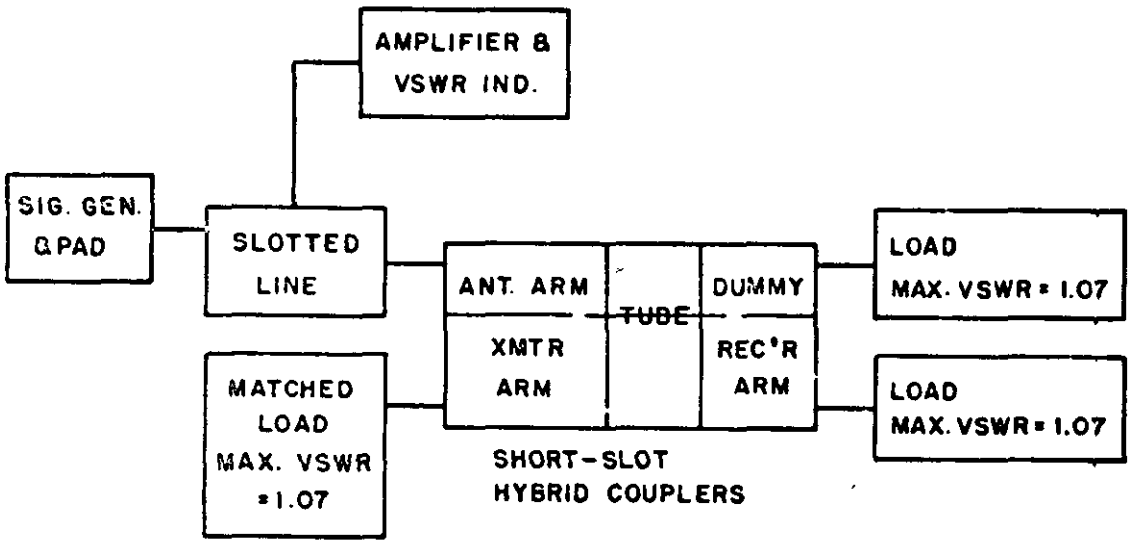
REF	DIMENSIONS
A ✱✱	1/8 X 45° APPROX.
B ✱✱	2.892
C ✱	1.446
D	8-32 TAP(M) HOLES E ... FIG.
E ✱✱	3 7/32
F ✱	.722
G ✱	1.000
H ✱	1-5/8
J ✱✱	1-1/8
K	1.272 ± .010
L ✱	1- 3/8 MAX.
M ✱✱	.250 NOM.

NOTE: EXHAUST STEM NOT TO EXTEND BEYOND FLANGES.

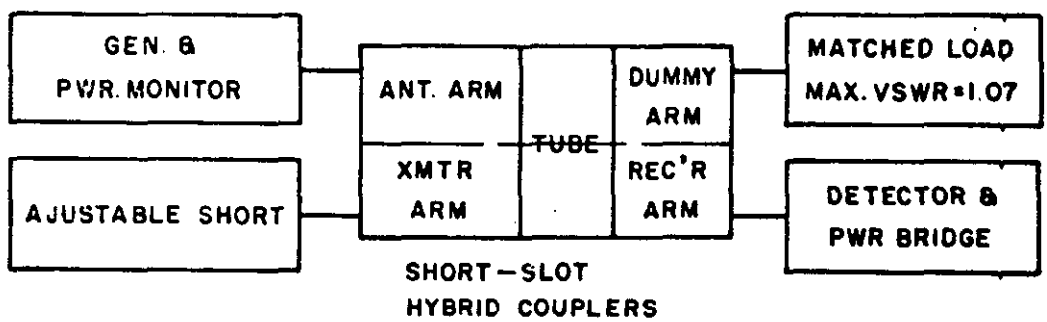


INPUT AND OUTPUT FLANGES
FIG. 2

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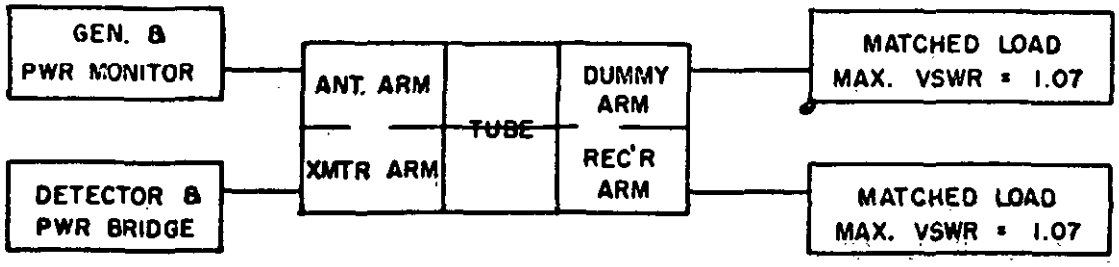
TEST CIRCUIT A



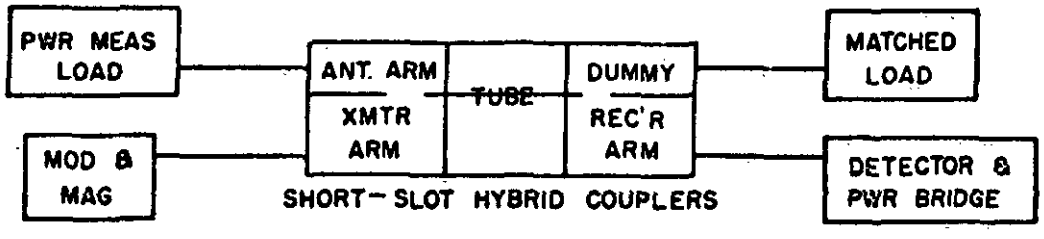
TEST CIRCUIT B

TEST CIRCUITS A AND B
FIG. 3

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TEST CIRCUIT C

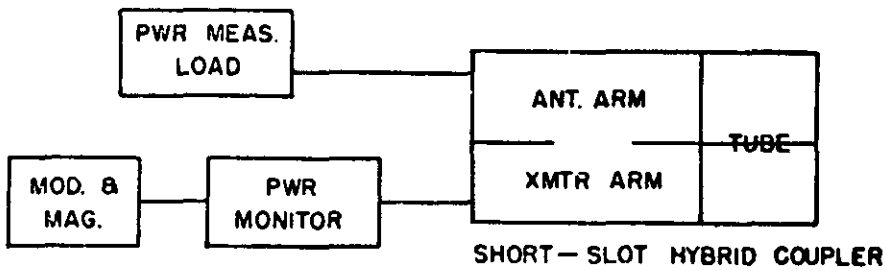


TEST CIRCUIT D

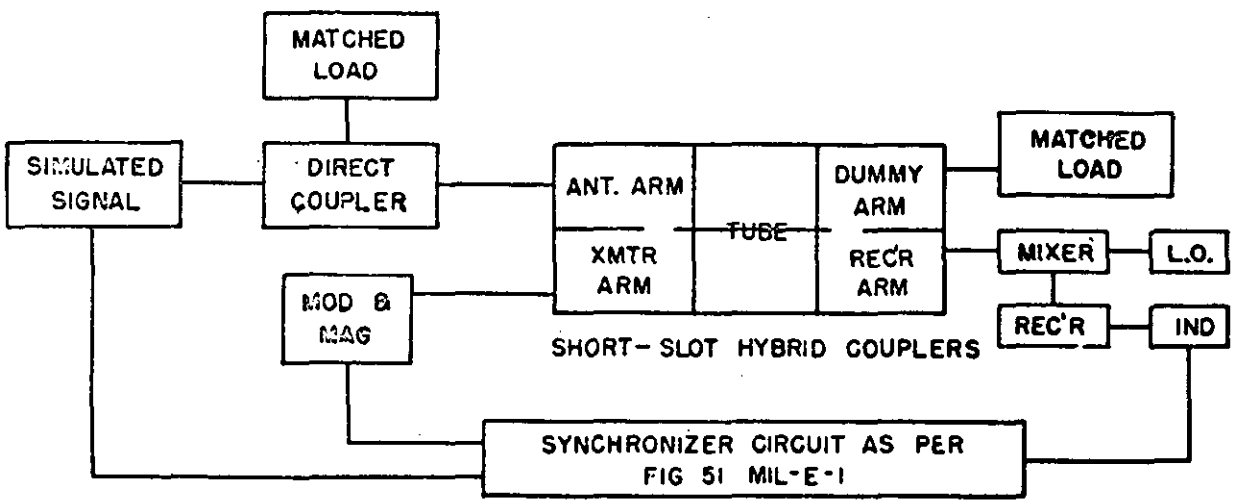
TEST CIRCUITS C AND D

FIG. 4

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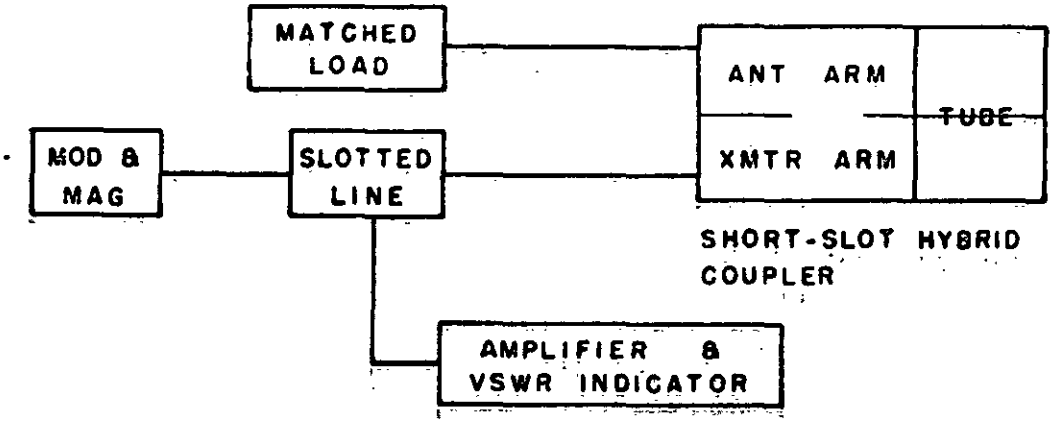
TEST CIRCUIT E



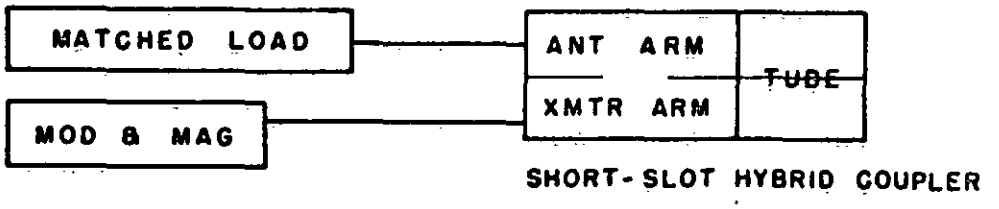
TEST CIRCUIT F

TEST CIRCUITS E AND F
FIG. 5

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TEST CIRCUIT G



TEST CIRCUIT H

TEST CIRCUITS G AND H
FIG 6

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